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Disclosures

Disclosure forms are available with the article online.

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Cardiac Tamponade, Mesenteric Ischemia, and Macrophage Activation Syndrome as the Initial Manifestation of Systemic Lupus Erythematosus: A Case Report

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Keywords

Systemic lupus erythematosus, Cardiac tamponade, Mesenteric ischemia, Vasculitis, Macrophage activation syndrome

Abstract

A 36-year-old man presented with dyspnea requiring emergent pericardiocentesis for cardiac tamponade (700 mL). On hospital day 19, he developed acute mesenteric ischemia; exploratory laparotomy revealed vasculitic vessel wall thickening without retrievable thrombus on embolectomy. Postoperatively, he developed macrophage activation syndrome with ferritin >16 000 ng/mL and transaminitis >3000 U/L. Systemic lupus erythematosus was diagnosed (ACR/EULAR score 21), representing the first reported case to our knowledge of simultaneous cardiac tamponade, mesenteric vasculitis, and macrophage activation syndrome at SLE onset. Treatment with cyclophosphamide and rituximab achieved remission. The surgical finding of vasculitis rather than thrombosis proved critical in guiding immunosuppressive therapy.

Background

Systemic lupus erythematosus (SLE) is a chronic multisystem autoimmune disease characterized by diverse clinical manifestations (1). Severe manifestations requiring hospitalization occur in 20% to 30% of patients at disease onset (2). Severe SLE is defined by SLEDAI-2K scores greater than or equal to 12 or BILAG grade A involvement (3, 4). Cardiac tamponade, large-vessel vasculitis, and macrophage activation syndrome (MAS) each occur in less than 3% of new-onset SLE cases; their simultaneous presentation is unprecedented.

We present the case of a 36-year-old man with cardiac tamponade, acute mesenteric vasculitis, and MAS as the initial manifestation of SLE. Systematic literature review confirmed this as the first reported case of this catastrophic triad to our knowledge. This case demonstrates critical diagnostic challenges and the importance of surgical findings in distinguishing vasculitis from thrombosis.

Case Report

A 36-year-old man with childhood asthma presented with dyspnea. Tachycardia (heart rate, 108 beats/min) and tachypnea (respiratory rate, 25 breaths/min) were disproportionate to oxygen saturation (99% on room air). Point-of-care transthoracic echocardiography revealed large pericardial effusion with right ventricular diastolic collapse (Figure 1). Emergent pericardiocentesis drained 700 mL of serosanguineous fluid.

Autoimmune work-up revealed anti-dsDNA antibody, 28 IU/mL (reference range, <10), antinuclear antibody (ANA) 1:160 (homogeneous/speckled), low C3 level (41 mg/dL), and C4 level (11 mg/dL). The patient met ACR/EULAR 2019 criteria for SLE with 21 points (threshold, ≥10), including pericarditis (6 points), anti-dsDNA (6 points), low complement (4 points), antiphospholipid antibody test (2 points), fever (2 points), and arthritis (1 point). Cardiac magnetic resonance imaging confirmed myopericarditis with reduced ejection fraction (31%), elevated T2 signal, and subepicardial late gadolinium enhancement. Left-sided thoracentesis

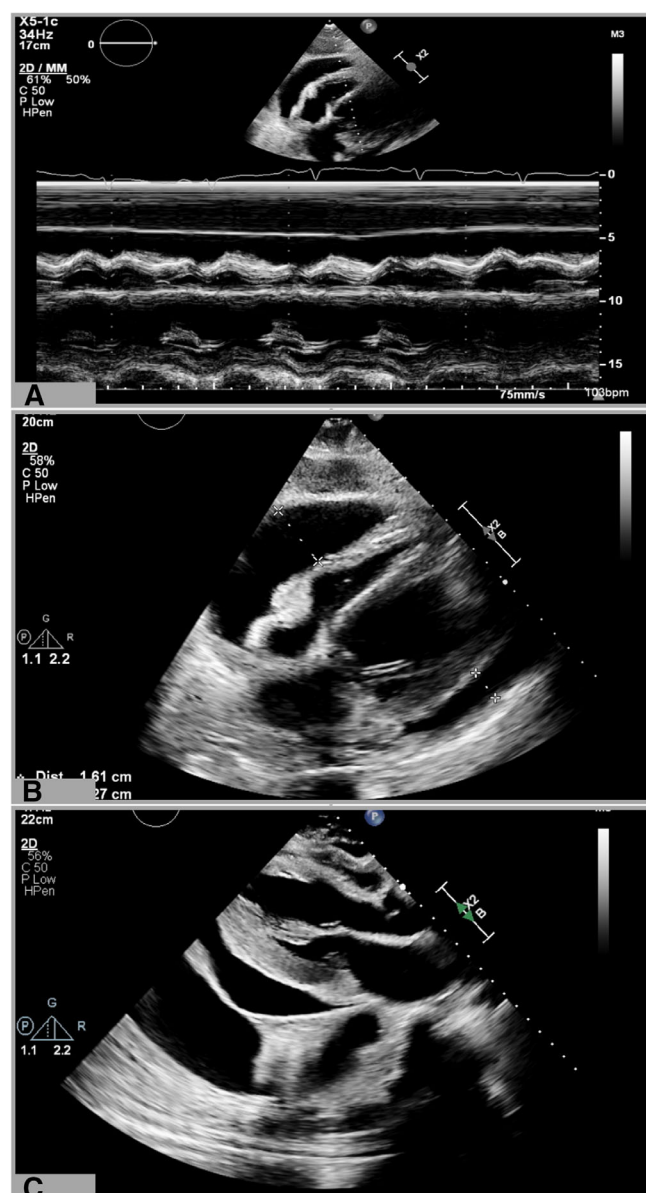


Figure 1. Transthoracic echocardiography demonstrating cardiac tamponade. Echocardiographic findings on hospital day 1 at initial presentation. A. Parasternal long-axis view (top) with corresponding M-mode echocardiogram (bottom) demonstrates large pericardial effusion (echo-free space) surrounding the heart. B. Apical 4-chamber view shows circumferential pericardial effusion measuring 3.6 cm anteriorly with compression of cardiac chambers. C. Subcostal 4-chamber view confirms large pericardial effusion measuring 3.0 cm with hemodynamic compromise. Point-of-care ultrasound identified right ventricular diastolic collapse, pathognomonic for cardiac tamponade. Emergent pericardiocentesis drained 700 mL serosanguineous fluid with immediate hemodynamic improvement.

yielded 750 mL of monocyte-predominant fluid; combined with elevated B-type natriuretic peptide level (peaking at 4144 pg/mL) and troponin (1837 ng/mL), findings indicated heart failure effusion with superimposed lupus serositis.

On hospital day 19, acute abdominal pain developed. Computed tomography angiography revealed superior mesenteric artery occlusion with diffuse wall thickening of the infrarenal aorta and bilateral renal arteries, suggesting large-vessel vasculitis (3)

(Figure 2). Magnetic resonance angiography confirmed celiac narrowing, superior mesenteric artery (SMA) stenosis, bilateral renal artery stenosis, and bilateral iliac narrowing.

Cardiovascular risk assessment showed low atherosclerotic burden (total cholesterol level, 120 mg/dL, nonsmoker, no diabetes/hypertension). Antiphospholipid antibody testing revealed isolated transient lupus anticoagulant (dRVVT 1.24, normalizing to 0.93 by day 27) with negative anticardiolipin and β 2-glycoprotein antibodies—a “single positive” profile.

Exploratory laparotomy revealed a pale SMA without visible thrombus. Fogarty catheter embolectomy retrieved no clot material despite aggressive attempts. The surgical team observed a thickened, inflamed vessel wall rather than discrete thrombotic occlusion. No bowel necrosis was identified. This surgical finding provided evidence supporting vasculitis over thromboembolism.

Postoperatively, fever (38.3 °C) and multiorgan dysfunction developed. Laboratory evaluation revealed extreme hyperferritinemia (16 104 ng/mL on day 26), severe transaminitis (aspartate aminotransferase/alanine aminotransferase >3000 U/L), elevated triglyceride level (176 mg/dL), elevated soluble IL-2 receptor level (1280 pg/mL), hepatomegaly, and anemia. The patient’s fibrinogen level was elevated (459 mg/dL), excluding consumptive coagulopathy. His H-Score was calculated at 136 points (88% probability of hemophagocytic lymphohistiocytosis/macrophage activation syndrome) based on extreme hyperferritinemia, >6000 ng/mL (50 points), hepatomegaly (23 points), elevated triglyceride level (44 points), and elevated aspartate aminotransferase (19 points). Bone marrow biopsy showed no hemophagocytosis. However, clinical diagnosis was established based on extreme hyperferritinemia (>10 000 ng/mL has 96% specificity for hemophagocytic lymphohistiocytosis/macrophage activation syndrome) and a temporal relationship to surgical stress.

High-dose corticosteroids were deferred pending tuberculosis (TB) exclusion (3 indeterminate QuantiFERON assays, pulmonary nodules, TB exposure history). His comprehensive infectious work-up was negative (respiratory viral panel, Epstein–Barr virus, cytomegalovirus, parvovirus, HIV, hepatitis panel, acid-fast *Bacillus* cultures, MTB PCR, T-SPOT). Pulse methylprednisolone (1000 mg IV daily $\times$ 3) was initiated on day 17. Bilateral papilledema developed, resolving by week 8 with immunosuppression.

On day 25, cyclophosphamide 1000 mg IV and rituximab were administered, achieving rapid response: the patient’s ferritin level decreased 70% within 72 hours (16 104 $\rightarrow$ 4962 ng/mL) and his transaminase level normalized within 1 week. A second cyclophosphamide dose was given on day 39. The patient was discharged on day 45 on prednisone 50 mg, heart failure medications, and rivaroxaban for residual thrombotic risk. Serial laboratory values are shown in Table 1, which contains key laboratory parameters demonstrating disease progression

Table 1. Serial Laboratory Values During Hospitalization

Parameter	Reference Range	Day 1, 9/5	Day 7, 9/12	Day 14, 9/18	Day 19, 9/23	Day 26, 9/30	Day 28, 10/2	Day 39, 10/14	Day 46, 10/20
Hematology									
Leukocyte count, K/ μ L	4.0–11.0	14.7	16.3	16.1	22.8	22.2	—	11.7	7.2
Hemoglobin level, g/dL	13.0–17.0	12.5	13.1	11.2	10.2	9.7	—	10.3	9.7
Platelet count, K/ μ L	150–400	708	784	698	720	763	—	490	345
Inflammatory									
ESR, mm/h	<20	62	53	84	78	43	33	12	8
CRP level, mg/L	<10	63.3	12.4	72.7	28.2	52	24.1	5.3	5.0
Ferritin level, ng/mL	<300	277	—	—	—	16 104	4962	234	210
sIL-2R level, pg/mL	<1000	—	—	—	—	1280	—	—	—
Hepatic									
AST level, U/L	<40	15	18	—	362	3305	—	19	12
ALT level, U/L	<40	<9	30	—	238	3613	—	68	28
Albumin level, g/dL	3.5–5.5	2.6	2.4	2.6	2.4	2.7	—	2.9	2.5
Total bilirubin level, mg/dL	0.1–1.2	—	0.5	—	0.3	0.8	—	0.8	0.6
Coagulation									
Fibrinogen level, mg/dL	200–400	—	—	—	—	459	415	—	—
Renal									
Creatinine level, mg/dL	0.7–1.3	0.92	0.92	0.86	1.89	1.33	—	0.73	0.80
BUN level, mg/dL	7–20	8	16	15	25	63	—	25	24
Protein/Cr ratio, mg/g	<200	—	102	819	—	348	—	—	—
Lipid									
Triglyceride level, mg/dL	<150	—	—	75	—	176	151	—	70
Immunologic									
C3 level, mg/dL	90–180	41	42	—	—	53	—	51	50
C4 level, mg/dL	10–40	11	18	—	—	21	—	23	23
Anti-dsDNA level, IU/mL	<10	28	19	—	—	9	—	5	4
Cardiac									
BNP level, pg/mL	<100	423	873	—	1479	4144	—	2608	—
Troponin I level, ng/mL	<26	33	353	—	1837	—	—	79	—
Other									
Lactate level, mmol/L	0.5–2.0	1.4	—	2.8	7.7	2.0	—	—	—

Bold values indicate life-threatening abnormalities. Em dash (—) indicates test not performed on that day.
ALT = alanine aminotransferase; AST = aspartate aminotransferase; BNP = B-type natriuretic peptide; BUN = blood urea nitrogen; CRP = C-reactive protein; ESR = erythrocyte sedimentation rate; dsDNA = double-stranded DNA; MAS = macrophage activation syndrome; Protein/Cr = protein-to-creatinine ratio; sIL-2R = soluble interleukin-2 receptor.

from initial presentation through MAS and subsequent response to immunosuppression. Ferritin peaked at 16 104 ng/mL on day 26 (>10 000 ng/mL has 96% specificity for HLH/MAS), declining 70% within 72 hours of cyclophosphamide and rituximab administration. Transaminases (AST/ALT) exceeded 3000 U/L at MAS peak, normalizing within 1 week of treatment. BNP progression reflects cardiac dysfunction from myopericarditis, peaking at 4144 pg/mL. A timeline of key events in the patient’s treatment is presented in Table 2. The complete treatment timeline with clinical decision-making rationale is detailed in Table 3.

Discussion

This case represents the first reported simultaneous presentation to our knowledge of cardiac tamponade, large-vessel mesenteric vasculitis, and macrophage activation syndrome as the initial manifestation of SLE. The patient’s SLEDAI-2K score of 25 at disease peak reflects severe multiorgan

involvement, driven by systemic vasculitis (8 points), arthritis (4 points), proteinuria (4 points), and serositis (4 points). A systematic literature review of 3580 articles identified 156 cases of these complications individually at SLE onset and 55 paired cases but no prior reports of this catastrophic triad

Table 2. Timeline of Key Events in the Patient’s Treatment

Day 1: Admission, pericardiocentesis (700 mL)
Day 7: Systemic lupus erythematosus diagnosis (ACR/EULAR score 21)
Day 14: Pericardial window procedure
Day 19: Exploratory laparotomy for mesenteric ischemia
Day 25: MAS diagnosis (H-Score 136); cyclophosphamide, 1000 mg + rituximab initiated
Day 28: Early response to immunosuppression
Day 39: Second cyclophosphamide dose (1000 mg)
Day 45: Hospital discharge

ACR/EULAR = American College of Rheumatology/European Alliance of Associations for Rheumatology; MAS = macrophage activation syndrome.

Table 3. Treatment Timeline with Clinical Decision-Making

Hospital Days	Dates	Treatment	Clinical Indication/Reasoning	Key Laboratory/Clinical Markers	Clinical Response
Initial presentation and diagnosis					
1	9/5	Solumedrol 40 mg IV q12h CTX + Doxycycline Emergent pericardiocentesis	Presumed asthma exacerbation CAP coverage Cardiac tamponade	Leukocyte count 14.7 BNP 423 Echo: tamponade physiology	Hemodynamically stable post-drainage
2–6	9/6–9/10	Continue solumedrol 40 mg IV q12h Antibiotics stopped Day 3	Autoimmune workup initiated ANA/dsDNA sent	ANA positive Anti-dsDNA 28 (positive) C3 41, C4 11 (both low)	Pericardial effusion re-accumulating EF decreased 50%→40%
7–8	9/11–9/12	Continue solumedrol 40 mg IV q12h Thoracentesis (750 cc L pleural)	SLE diagnosed (ACR score 21) Rheumatology consulted	ESR 53, CRP 12.4 Proteinuria 240 mg/g	Transferred to NYULIH for rheumatologic workup
Immunosuppression held—TB evaluation					
9–16	9/13–9/20	NO high-dose steroids Low-dose supportive care only Colchicine, ibuprofen	Indeterminate QuantiFERON ×3 Pulmonary nodules on CT TB exposure in Guyana ID: defer steroids pending TB r/o	QuantiFERON indeterminate (9/5, 9/7, 9/11) CT: 4 mm nodules RLL Awaiting AFB, MTB PCR, fungal panel	Pericardial window 9/18 for re-accumulation Cardiac MRI 9/13: LVEF 31%, myopericarditis
Pulse steroids for CNS lupus					
17–19	9/22–9/24	Methylprednisolone 1000 mg IV daily ×3 Colchicine, carvedilol	Bilateral papilledema (Day 10) MRA: cerebral vasculopathy Infectious workup negative	MRI/MRA brain: intracranial stenosis Ophthalmic: disc edema, APD bilaterally T-SPOT pending	Underwent laparotomy Day 19 for acute abdomen
Catastrophic vasculitis					
19–24	9/23–9/29	Solumedrol 30–40 mg IV BID (variable) Laparotomy 9/23: SMA embolectomy (negative) Re-exploration 9/25: No bowel necrosis Heparin gtt initiated	Acute mesenteric ischemia CTA: SMA occlusion + vessel wall thickening Surgical finding: vasculitis, not thrombus	Lactic acid 7.3 Negative embolectomy (no clot retrieved) CTA: aortic/renal artery wall thickening	Post-op deterioration High fevers developing LFTs rising
Macrophage activation syndrome					
25–26	9/30–10/1	Cyclophosphamide 1000 mg IV Rituximab (dose not specified) Decadron 18.3 mg IV daily (10 mg/m ²) Bactrim (PJP prophylaxis)	MAS diagnosed clinically Ferritin >16 000 = 96% specific for HLH H-Score 136 (88% probability)	Ferritin 16 104 (Day 26) AST/ALT >3000 Triglycerides 176 Hepatomegaly on US	Ferritin ↓ to 4962 by Day 28 Transaminases improving
Steroid transition					
27–28	10/2–10/3	Continue dexamethasone 18.3 mg daily	MAS improving	Ferritin continuing to decline LFTs normalizing	Clinical improvement
29–38	10/4–10/13	Switched to methylprednisolone 80 mg IV daily Taper to 60 mg by Day 33	Transition from HLH protocol Systematic taper	LFTs normalized Ferritin downtrending Inflammatory markers ↓	Proteinuria resolved (Day 40) Weight gain, improving strength
Second CYC dose					
39	10/14	Cyclophosphamide #2: 1000 mg IV Solumedrol 60 mg IV daily	Planned induction per protocol	Labs stable	Continued improvement
40–44	10/15–10/19	Taper to 50 mg IV→ Transition to PO prednisone 50 mg	Preparing for discharge	Repeat echo: mod pericardial effusion (stable) BNP 2608 (down from 4144)	Deemed stable for discharge

(continued)

Table 3—Continued

Hospital Days	Dates	Treatment	Clinical Indication/Reasoning	Key Laboratory/Clinical Markers	Clinical Response
Discharge					
45	10/20	Discharge medications: Prednisone 50 mg PO daily Losartan 50 mg Carvedilol 12.5 mg BID Spironolactone 25 mg Rivaroxaban 15 mg BID→ 20 mg daily Sulfamethoxazole/ trimethoprim SS daily Colchicine 0.6 mg daily	Multispecialty f/u arranged: Rheumatology Cardiology/HF Vascular surgery Neurology Ophthalmology Infectious disease	Off IV therapy Hemodynamically stable Ambulating with assistance	Persistent mononeuritis multiplex (foot drop) EF 20% (severely reduced)
Outpatient continuation					
53 (D+8)	10/28	Cyclophosphamide #3: 1000 mg IV Prednisone taper to 40 mg	Modified Euro-Lupus protocol Monthly dosing after initial 2 doses	Leukocyte count 12.3 (reactive leukocytosis) No infection	Energy improved Able to walk independently
68 (D+23)	11/12	Prednisone taper to 30 mg No CYC (q monthly)	Steroid-sparing strategy	Papilledema RESOLVED (ophthalmic exam) Visual acuity normalized	Significant functional improvement
80 (D+35)	11/24	Cyclophosphamide #4: 1000 mg IV Prednisone 30 mg	Maintenance dosing	Leukocyte count 14.9, Hgb improving Heme cleared (reactive process)	Nerve pain/weakness improving Walking without assistance
91 (D+46)	12/5	Prednisone taper to 25 mg	Continued slow taper	Leukocyte count stable No infectious symptoms	“Significantly improved” Able to ambulate freely
109 (D+64)	12/23	Cyclophosphamide #5: 1000 mg IV (planned) CYC STOPPED after this dose per rheumatology team Prednisone 25 mg	Total 5 doses vs standard 6 Decision based on excellent response	Disease activity controlled Target organ damage resolved	Plan for continued slow steroid taper Consider maintenance agent once EF improves

Comprehensive timeline of therapeutic interventions demonstrating individualized treatment approach balancing disease activity against infection risk. High-dose corticosteroids were appropriately deferred for 12 days pending tuberculosis exclusion given three indeterminate QuantiFERON assays, pulmonary nodules, and TB exposure history. Pulse methylprednisolone was initiated on day 17 after comprehensive infectious workup showed no evidence of active infection. Cyclophosphamide 1000 mg IV plus rituximab on day 25 achieved rapid biochemical response with ferritin declining from 16,104 to 4962 ng/mL within 72 hours and transaminase normalization within one week.

AFB = acid-fast bacilli; ANA = antinuclear antibody; APD = afferent pupillary defect; BNP = brain natriuretic peptide; CT = computed tomography; CTA = computed tomography angiography; CTC = circulating tumor cell; CTX = ceftriaxone; CYC = cyclophosphamide; dsDNA = double-stranded DNA; ESR = erythrocyte sedimentation rate; MAS = macrophage activation syndrome; HLH = hemophagocytic lymphohistiocytosis; EF = ejection fraction; HF = heart failure; HLH = hemophagocytic lymphohistiocytosis; LFT = liver function test; LVEF = left ventricular ejection fraction; MRA = magnetic resonance angiography; MRI = magnetic resonance imaging; MTB = *Mycobacterium tuberculosis*; NYULIH = New York University Long Island Hospital; PCR = polymerase chain reaction; PJP = *Pneumocystis jirovecii* pneumonia; RLL = right lower lobe; r/o = rule out; SLE = systemic lupus erythematosus; SMA = superior mesenteric artery; TB = tuberculosis.

confirming the unprecedented nature of this presentation (Table 4). The review methodology was as follows. PubMed and Google Scholar searches were conducted for the years 1990 to 2024 using the terms “systemic lupus erythematosus,” “cardiac tamponade,” “mesenteric vasculitis,” “macrophage activation syndrome,” and “new-onset” OR “initial presentation.” The inclusion criteria were English-language case reports or series documenting these complications at SLE diagnosis (within 6 months of meeting ACR/EULAR criteria). The key studies presented in Table 4 are organized by complication type with sample sizes, demographics, and clinical outcomes.

The critical diagnostic challenge in this case was distinguishing lupus mesenteric vasculitis from thrombotic antiphospholipid syndrome. The isolated SMA occlusion with transient lupus anticoagulant initially raised concern for thrombosis. While antiphospholipid syndrome (APS) cannot be definitively

excluded, several lines of evidence supported primary vasculitis: “single positive” transient lupus anticoagulant without a triple-positive pattern typical of high-risk APS; surgical finding of no retrievable thrombus on Fogarty embolectomy despite aggressive attempts; radiographic vessel wall thickening on CT/MRA demonstrating systemic vasculitic process; and fibrinogen level elevation (459 mg/dL) rather than consumption. This distinction dictated cyclophosphamide and corticosteroids over anticoagulation alone. Excellent response to immunosuppression supports vasculitis diagnosis, though therapeutic anticoagulation was maintained for residual thrombotic risk.

Large-vessel lupus vasculitis, though classically affecting small–medium vessels, is increasingly recognized (5). The surgical finding of thickened, inflamed vessel walls without discrete thrombus was critical in establishing this diagnosis.

Table 4. Literature Review of Key Studies on Life-Threatening SLE Complications at Disease Onset

Author(s)/Year	Study Type	Patient Cohort (n)	Key Findings Relevant to Complications at SLE Onset
Cardiac tamponade at SLE onset			
Cohort studies			
Goswami et al. (2018) (6)	Retrospective cohort	n = 409 (India) (24 with tamponade)	12/24 (50%) had tamponade as the presenting feature. Predictors: Effusion > 20 mm (OR, 93.2), Pleuritis (OR, 44.5) Low C4 was the strongest serologic predictor.
Kahl (1992) (7)	Cohort study	n = 395 (USA)	40% of all tamponade cases in the cohort presented at disease onset.
Pediatric cases			
Maharaj and Chang (2015) (8)	Systematic review	n = 13 cases (Total, 15 cases from 1972–2024)	92.3% female. Ages 3–17 y (mean 10–11). All required emergent pericardiocentesis (500–1500 mL). 38.5% had co-existing pleural effusion.
Saz et al. (2010) (9)	Case report	n = 1	Youngest reported case: 3-year-old female patient.
Adult cases			
General (1992–2024)	Case reports/series	n = 40+	Ages 22–73 y. 85% female.
Mesenteric vasculitis (LMV) at SLE onset			
Cohort studies			
Yuan et al. (2014) (10)	Retrospective cohort	n = 3823 (China) (97 LMV cases)	2.5% prevalence. 47/97 presented with LMV as the initial SLE manifestation. Mortality 13.4%. High-dose cyclophosphamide was protective (OR, 0.220). Risk factors for death: leukopenia, hypoalbuminemia.
Pediatric cases			
Zhu et al. (2023) (11)	Case Series	n = 11 (China) (All new-onset SLE)	Mean age, 12.55 y (9F/2M). 100% had acute abdominal pain; 90.9% had concurrent renal involvement. 0% required surgery, 0% mortality. 100% received cyclophosphamide; 90.9% received methylprednisolone pulse.
Adult cases			
Alshehri (2020) (12)	Case report	n = 1	30-year-old woman. CT + target sign. Responded to prednisolone within 2 weeks.
Nozari and Divsalar (2014) (13)	Case Report	n = 1	Fatal case: 34-year-old woman underwent diagnostic laparoscopy, developed DIC, and died postoperatively, emphasizing the risk of surgical intervention.
Macrophage activation syndrome (MAS)/HLH at SLE onset			
Cohort studies			
Abdirakhmanova et al. (2021) (14)	Systematic review	n = 46 (pediatric)	74.3% (26/35) had MAS before pSLE diagnosis. Mortality, 13.0% (6/46).
Borgia et al. (2018) (15)	Cohort study	n = 403 (cSLE)	9% prevalence. 68% had MAS and SLE diagnosed concomitantly. Hospital mortality OR, 64.5 in the MAS group.
Notable case reports			
Alkoht et al. (2017) (16)	Case report	n = 4 (pediatric)	Closest case to a triad pair: 4-year-old female patient (youngest) with MAS (ferritin level, 12 000) plus pericardial effusion requiring drainage. Required ICU, methylprednisolone, and cyclophosphamide.
Thornton et al. (2019) (17)	Case report	n = 1 (adult)	Extreme hyperferritinemia: 43-year-old man. Ferritin level, >100 000 ng/mL. Required ICU/intubation/dialysis. Survived with methylprednisolone, anakinra, cyclophosphamide.
Thornton et al. (2019) (17)	Case Report	n = 1 (adult)	Oldest case: 73-year-old man. H-Score, 245 (99.1% HLH probability). Ferritin level, >8000. Achieved remission with methylprednisolone and hydroxychloroquine.
Bhowmick et al. (2025) (18)	Case report + review	n = 1 (adolescent)	19-year-old woman with MAS, hypoplastic marrow, and orbital myositis as initial SLE presentation. Demonstrates that MAS can present with atypical features at disease onset.

ACR = American College of Rheumatology; DIC = disseminated intravascular coagulation; EULAR = European Alliance of Associations for Rheumatology; HLH = hemophagocytic lymphohistiocytosis; ICU = intensive care unit; MAS = macrophage activation syndrome; OR = odds ratio; SLE = systemic lupus erythematosus.

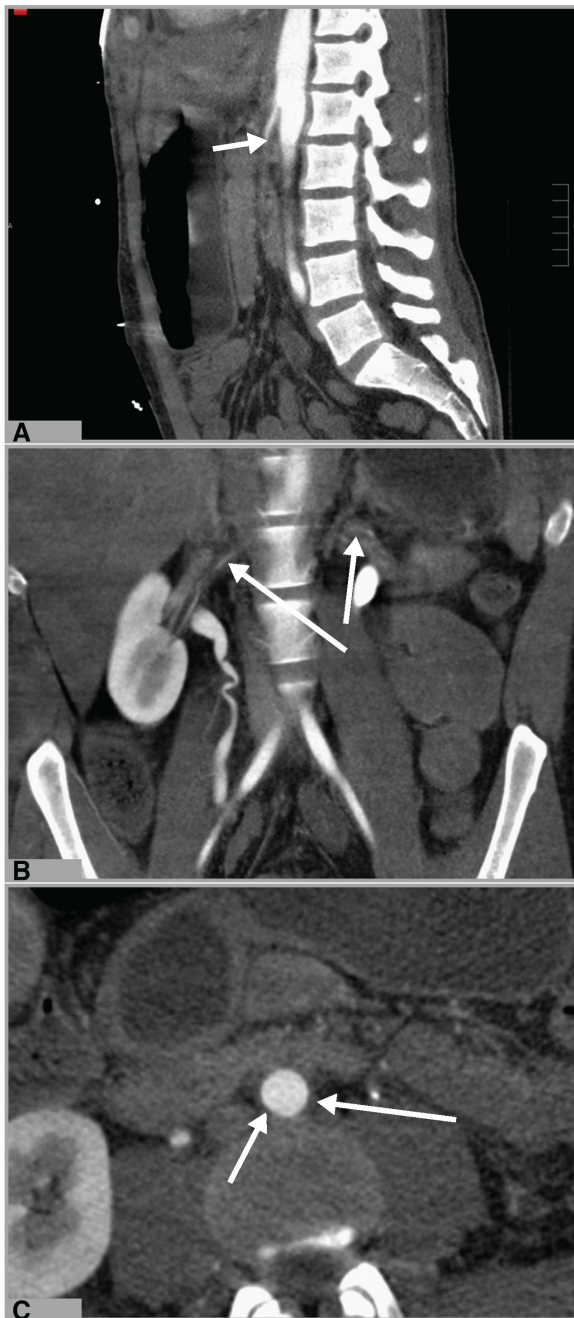


Figure 2. Computed tomography angiography demonstrating mesenteric vasculitis. Contrast-enhanced computed tomography angiography of the abdomen and pelvis obtained on hospital day 19 demonstrating diffuse large-vessel vasculitis. A. Sagittal reconstruction shows infrarenal aortic wall thickening (arrow) extending to the aortic bifurcation. B. Coronal reconstruction demonstrates SMA occlusion with circumferential vessel wall thickening (arrows) affecting the infrarenal aorta and SMA origin, consistent with vasculitic process rather than discrete intraluminal thrombus. Kidneys (bright structures laterally) show normal contrast enhancement. C. Axial view at the level of the SMA origin shows cross-sectional vessel wall thickening (arrows) with narrowed lumen. Additional imaging (not shown) revealed bilateral renal artery stenosis and bilateral common iliac artery involvement. Subsequent exploratory laparotomy with Fogarty catheter embolectomy retrieved no thrombus despite aggressive attempts, providing surgical confirmation of vasculitic rather than embolic/thrombotic cause. This intraoperative finding was critical in guiding treatment toward aggressive immunosuppression (cyclophosphamide and rituximab) rather than anticoagulation alone. SMA = superior mesenteric artery.

Macrophage activation syndrome at SLE onset is rare. Our patient's H-Score of 136 (88% probability) was driven by extreme hyperferritinemia (16 104 ng/mL), severe transaminitis (>3000 U/L), hepatomegaly, and hypertriglyceridemia. Ferritin greater than 10 000 ng/mL has 96% specificity for HLH/MAS, supporting clinical diagnosis despite negative bone marrow. The temporal relationship to laparotomy suggests a "2-hit" mechanism: active lupus plus surgical stress precipitating cytokine storm (18).

High-dose corticosteroids were appropriately deferred 12 days for tuberculosis exclusion given indeterminate testing and exposure history. Cyclophosphamide and rituximab achieved rapid disease control: ferritin declined 70% within 72 hours and transaminases normalized within 1 week. This case demonstrates that catastrophic SLE can present at disease onset and that surgical findings may provide critical diagnostic information distinguishing vasculitis from thrombosis, fundamentally altering the therapeutic approach.

While we present this as the first reported simultaneous triad at SLE onset to our knowledge, our literature search was limited to English-language publications. Antiphospholipid syndrome cannot be definitively excluded given single positive transient lupus anticoagulant without 12-week confirmatory testing. Bone marrow biopsy was negative for hemophagocytosis, though this occurs in up to 40% of MAS cases. Visual evoked potentials were not performed during acute hospitalization due to competing clinical priorities.

This case represents the first reported simultaneous presentation to our knowledge of cardiac tamponade, mesenteric vasculitis, and MAS as the initial manifestation of SLE. The patient's SLEDAI-2K score of 25 reflects severe multiorgan involvement at disease onset, challenging the paradigm that catastrophic complications occur only in established disease.

The surgical finding of vasculitic vessel wall thickening without retrievable thrombus proved diagnostically critical, guiding treatment toward aggressive immunosuppression rather than anticoagulation. Despite initial steroid delay for tuberculosis exclusion, cyclophosphamide and rituximab achieved rapid disease control with normalization of inflammatory markers and functional recovery. Intraoperative findings in acute mesenteric ischemia may provide critical diagnostic information differentiating vasculitis from thrombosis, fundamentally altering the therapeutic approach.

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